

Sinian Zhang

📍 Minneapolis, Minnesota, US ✉ zhan9381@umn.edu
🌐 github.com/h7nian 🌐 h7nian.github.io 🎓 Google Scholar

EDUCATION

University of Minnesota, Twin Cities

Minneapolis, US

Ph.D. Student in Biostatistics

Sept 2024 – Jun 2029 (Expected)

Dissertation Advisors: Sandra Safo and Jue Hou (Division of Biostatistics and Health Data Science)
Also Mentored by: Rui Zhang (Division of Computational Health Sciences) and Ju Sun (Computer Science & Engineering)

Renmin University of China

Beijing, China

B.S. in Statistics

Sept 2020 – Jun 2024

RESEARCH INTERESTS

Statistically principled methods for **transferable and trustworthy machine learning and deep learning**, with applications spanning biomedical, imaging, and engineering domains.

- **Uncertainty Quantification & Trustworthy AI** – selective prediction and classification, risk-coverage optimization, uncertainty-aware decision making.
- **Agentic AI & LLM Systems** – multi-agent workflows for automated scientific data analysis, human-in-the-loop orchestration, retrieval-augmented reasoning over biomedical knowledge.
- **Transfer Learning & Causal Inference** – distributional transfer learning, federated causal estimation, high-dimensional confounding, reinforcement learning.
- **Health Informatics & Real-World Evidence** – electronic health record phenotyping, comparative effectiveness, biomedical knowledge graphs.

PUBLICATIONS

**Equal contribution.*

Peer-Reviewed Publications

1. Jingxin Wang*, **Sinian Zhang***, Yun Liang, Yibo Lin, Pengpeng Ren, Runsheng Wang, Weikang Qian. GUALO: A Generalizable Uncertainty-Aware AI Agent for Logic Optimization. *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2026.
Role: Co-designed the uncertainty-aware agent and led the uncertainty-quantification methodology and experiments.
2. Elzbieta Jodlowska-Siewert, Yunhui Chen, **Sinian Zhang**, Jia Li, Robert Dellavalle, Rui Zhang, Jue Hou. Antidiabetic Drug Associations With Heart Failure Outcomes: Real-World Evidence Study Using Electronic Health Records. *JMIR Diabetes* 11 (2026): e85083.
Role: Performed the electronic health record data processing.
3. Kaicheng Zhang*, **Sinian Zhang***, Doudou Zhou, Yidong Zhou. Wasserstein Transfer Learning. *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
Role: Co-developed the transfer-learning method and theory; led the simulation and real-data studies.
4. Yongkang Xiao, **Sinian Zhang**, Yi Dai, Huixue Zhou, Jue Hou, Jie Ding, Rui Zhang. DrKGC: Dynamic Subgraph Retrieval-Augmented LLMs for Knowledge Graph Completion across General and Biomedical Domains. *Findings of the Association for Computational Linguistics: EMNLP 2025*,

pages 16432–16445.

Role: Contributed to the retrieval-augmented framework design and experiments.

5. Feiqing Huang, Jue Hou, Ningxuan Zhou, Kimberly Greco, Chenyu Lin, Sara Morini Sweet, Jun Wen, Lechen Shen, Nicolas Gonzalez, **Sinian Zhang**, et al. Advancing the Use of Longitudinal Electronic Health Records: Tutorial for Uncovering Real-World Evidence in Chronic Disease Outcomes. *Journal of Medical Internet Research* 27 (2025): e71873.

Role: Contributed the longitudinal EHR data processing and curation.

6. Yongkang Xiao, **Sinian Zhang**, Huixue Zhou, Mingchen Li, Han Yang, Rui Zhang. FuseLinker: Leveraging LLM's Pre-trained Text Embeddings and Domain Knowledge to Enhance GNN-based Link Prediction on Biomedical Knowledge Graphs. *Journal of Biomedical Informatics* 158 (2024): 104730.

Role: Contributed to the GNN link-prediction methodology and experiments.

7. Jue Hou, Rachel Zhao, Jessica Gronsbell, Yucong Lin, Clara-Lea Bonzel, Qingyi Zeng, **Sinian Zhang**, et al. Generate Analysis-Ready Data for Real-world Evidence: Tutorial for Harnessing Electronic Health Records With Advanced Informatic Technologies. *Journal of Medical Internet Research* 25 (2023): e45662.

Role: Contributed to EHR data processing for generating analysis-ready datasets.

8. Shiyao Huang, Junliang Lyu, **Sinian Zhang**, Ruiying Tang, Huan Xiao, Yuanyuan Zhang, Xiaoling Lu. A Post-processing Machine Learning for Activity Recognition Challenge with OpenStreetMap Data. *Adjunct Proceedings of UbiComp/ISWC 2023*, pages 557–562.

Role: Contributed to the post-processing machine-learning method.

Preprints & Under Review

1. Jiajun Liang, Yue Liu, Doudou Zhou, **Sinian Zhang**, Junwei Lu. The Wreaths of KHAN: Uniform Graph Feature Selection with False Discovery Rate Control. *In revision*, *Biometrika*. arXiv preprint: [2403.12284](#), 2024.
2. Gaoxiang Luo, Yifan Wu, **Sinian Zhang**, Aryan Deshwal, Ju Sun. Aligning Language Models with Selective Prediction. *Under review*, Annual Conference on Neural Information Processing Systems (NeurIPS), 2026. arXiv preprint: [2607.03528](#), 2026.
3. **Sinian Zhang**^{*}, Chongwei Chen^{*}, Guanchen Li, Ju Sun. Ensemble Selective Classification. *Under review*, Annual Conference on Neural Information Processing Systems (NeurIPS), 2026.
4. Shruthi Venkatesh^{*}, **Sinian Zhang**^{*}, Wen Zhu, Michele Morris, Rocco Mercurio, et al. Predicting the Timing of First Sustained Cognitive Worsening in Alzheimer's Disease Using Real-World Clinical Data and Machine Learning. *Under review*, *npj Aging*. *medRxiv* preprint: [10.64898/2026.06.02.26354764](#), 2026.
5. **Sinian Zhang**^{*}, Kaicheng Zhang^{*}, Ziping Xu, Tianxi Cai, Doudou Zhou. Generalized Linear Markov Decision Process. arXiv preprint: [2506.00818](#), 2025.

Working Papers

1. **Sinian Zhang**, Chongwei Chen, Jiandong Chen, Lingjie Su, Ju Sun. Risk-Aligned Confidence for Selective Segmentation. *In preparation*, 2026.
2. **Sinian Zhang**, Sandra Safo. Supervised Multi-View Deep Learning with Exponential-Family Reconstruction, Feature Selection, and Fairness Assessment. *In preparation*, 2026.
3. Jue Hou^{*}, **Sinian Zhang**^{*}, Andrea Rotnitzky, James M. Robins, Tianxi Cai. Triply Robust Surrogate Assisted Semi-supervised Estimation of High-dimensional Linear Regression with Missing-at-Random Outcomes. *In preparation*, 2026.
4. **Sinian Zhang**, Yuanlin Feng, Jue Hou. Federated Adaptive Causal Estimation with High-Dimensional Covariates (FACE-HD). *In preparation*, 2026.
To be presented at the *7th ICSA-Canada Chapter Symposium*, McGill University, Montreal, Aug 2026.

5. **Sinian Zhang**, Jianfeng Wang, Thierry Chekouo. IntegMultiReg: An R Package for Integrative Bayesian Multi-Regression of Multi-Platform Biomarkers. *In preparation*, 2026.
Accompanying R package submitted to CRAN.

TECHNICAL SKILLS

Languages: R, Python, C, C++, SQL, L^AT_EX

Machine Learning: PyTorch, Hugging Face Transformers, scikit-learn

LLM & Agents: LangGraph, LLM APIs (Claude, Gemini, Ollama)

Tools: HPC/Slurm, Git, R package development (CRAN)

SOFTWARE

MultiViewAgent – Agentic AI system (Python, LangGraph)  2026

Multi-agent workflow that automates multiview biomedical data analysis over the Safo Lab multiview toolkit. Six specialized LLM agents (planner, recommender, executor, evaluator, supervisor, reporter) with supervisor rerouting, reflection loops, and human-in-the-loop review checkpoints; data-driven method selection across eleven multiview methods, with pluggable LLM backends and HPC deployment.

IntegMultiReg – R package (MCMC sampler in C) Submitted to CRAN, 2026

Integrative Bayesian multi-regression of multi-platform biomarkers, supporting continuous, binary and survival outcomes with missing platforms.

Khan – R package  2026

Sole author and maintainer. Companion package to The Wreaths of KHAN: FDR-controlled inference on subgraph features and persistent homology generators in Gaussian graphical and Ising models.

WaTL – R implementation  2025

Reference implementation of Wasserstein Transfer Learning (NeurIPS 2025), including the simulation and real-data analyses.

EnsembleSC – Python implementation  2025

Reference implementation of Ensemble Selective Classification (under review, NeurIPS 2026).

SelectiveSegmentation – Python implementation  2026

Reference implementation of Risk-Aligned Confidence for Selective Segmentation, including the reviewed experiment configurations and publication-facing result bundles.

NILE – Python and R implementations 2025

Natural language processing package for clinical narrative extraction; development guided as a mentor.

TALKS & PRESENTATIONS

Conference Abstracts

- Shruthi Venkatesh, **Sinian Zhang**, Oscar Lopez, Tianxi Cai, Jue Hou, Zongqi Xia. Predicting the Timing of Cognitive Decline in Alzheimer's Disease Using Real-world Clinical Data and Machine Learning (P10-12.009). *Neurology* 106 (11_Supplement_1), 2026.

Talks

- Federated Adaptive Causal Estimation with High-Dimensional Covariates (FACE-HD)**
7th ICSA-Canada Chapter Symposium, McGill University, Montreal, Canada Aug 2026

TEACHING & MENTORING

Shuheng Kong & Conglin Ruan – Master’s Students

May 2025 – Sept 2025

Guided development of Python and R versions of the NILE package for natural language processing.

Teaching Assistant, Data Science Practice

Beijing, China

Renmin University of China

Jan 2024 – Jun 2024

Teaching Assistant, Probability Theory

Beijing, China

Renmin University of China

Aug 2023 – Dec 2023

Brittany Wang – High School Student

Jun 2023 – Sept 2023

Supervised research using real-world data to support an acute ischemic stroke (AIS) study.

ACADEMIC SERVICE

Conference Reviewer: AAAI 2027, NeurIPS 2026, ICLR 2026

Journal Reviewer: Journal of Intelligent Medicine and Healthcare

HONORS & AWARDS

- **1st Place in the 7th Annual UMN Interdisciplinary Health Data Competition, \$3000** – University of Minnesota, Twin Cities, 2026
- **Dean’s PhD Scholars Award, \$5000** – University of Minnesota, Twin Cities, 2024
- **Scholarship for Academic Excellence** – Renmin University of China, 2023